

Class 9

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Class 9: Candy Mini-Project

2. Importing Candy data

```
#Importing Candy data
candy_file <- "candy-data.csv"
candy = read.csv (candy_file, row.names = 1)
#Viewing the first 6 rows of the dataset
head(candy)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	wafer
100 Grand	1	0	1			0	0		1
3 Musketeers	1	0	0			0	1		0
One dime	0	0	0			0	0		0
One quarter	0	0	0			0	0		0
Air Heads	0	1	0			0	0		0
Almond Joy	1	0	0			1	0		0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
#Viewing how many different candy types in this dataset
nrow(candy)
```

```
[1] 85
```

There are 85 different candy types in this data set

Q2. How many fruity candy types are in the dataset?

```
#Viewing sum of fruity candy types  
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types in this dataset.

2.2 What is your favorite candy?

```
#Viewing the percentage of people who prefer twix  
candy["Twix",]$winpercent
```

```
[1] 81.64291
```

```
#Loading dplyr for data functions  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
#Find the win percentage for twix  
candy |>  
  filter(row.names(candy)=="Twix") |>  
  select(winpercent)
```

```
      winpercent  
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Starburst", "winpercent"]
```

```
[1] 67.03763
```

The winpercent value of my favorite candy, Starburst, is 67.04.

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", "winpercent"]
```

```
[1] 76.7686
```

The winpercent value for Kit Kat is 76.77.

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", "winpercent"]
```

```
[1] 49.6535
```

The winpercent value for Tootsie Roll Snack Bars is 49.65.

```
library("skimr")  
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The winpercent column seems to be on a different scale to the majority of the other columns in the dataset. Most of columns in the data set range from 0 to 1 but the winpercent is ranging above 10.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

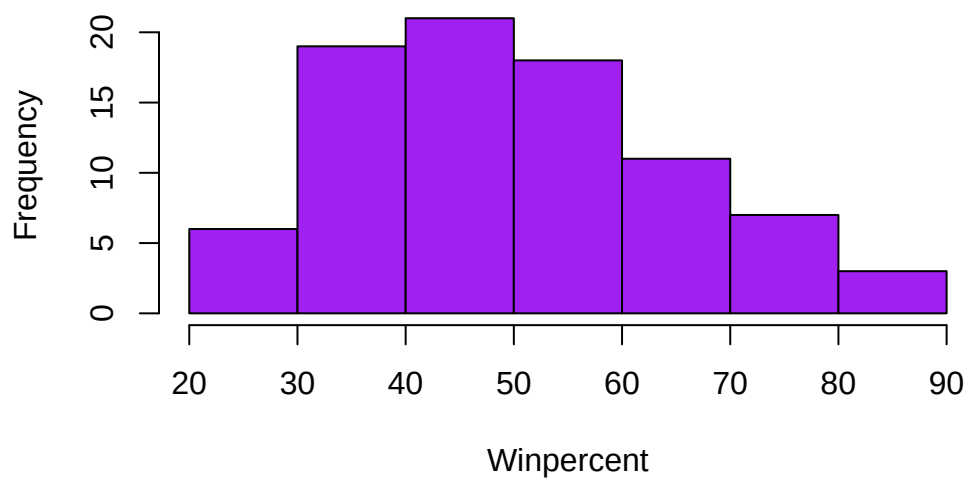
A zero in the candy\$chocolate column would represent that the candy does not have chocolate, but a 1 in this column would indicate that the candy does have a chocolate.

3. Exploratory Analysis

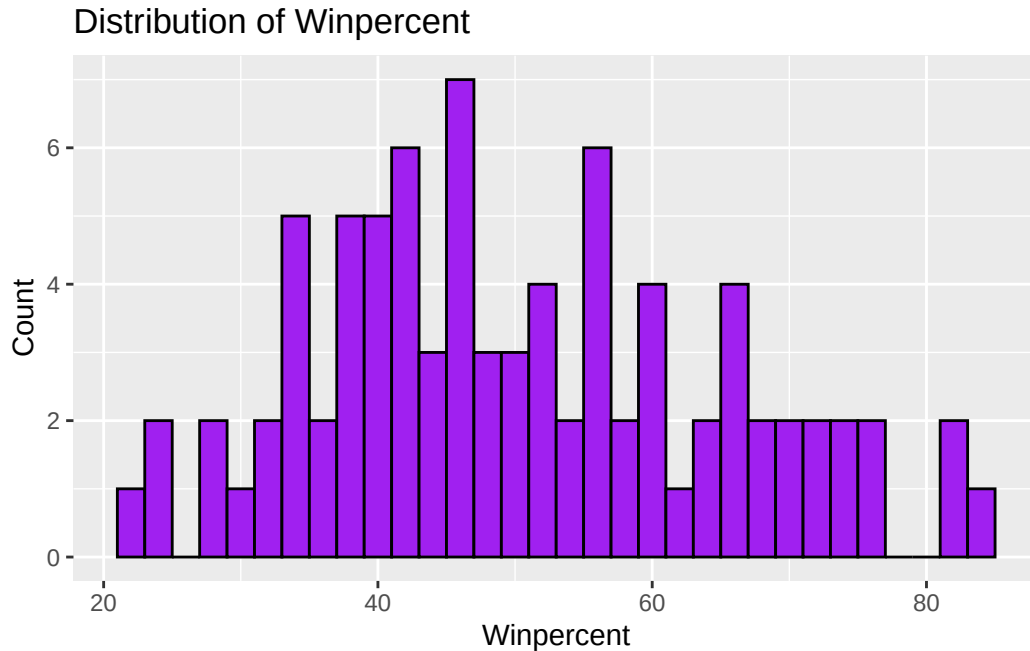
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
#Plotting a histogram of winpercent values using base R
hist(candy$winpercent,
     main = "Distribution of Winpercent",
     xlab = "Winpercent",
     col = "purple",
     border = "black")
```

Distribution of Winpercent



```
#Loading in ggplot2
library(ggplot2)
#Plotting a histogram of winpercent values using ggplot2
ggplot(candy, aes(x = winpercent)) +
  geom_histogram(binwidth = 2, fill = "purple", color = "black") +
  labs(title = "Distribution of Winpercent",
       x = "Winpercent",
       y = "Count")
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution of winpercent values is not symmetrical since it looks slightly right skewed, and there is a longer tail toward the higher winpercent values.

Q10. Is the center of the distribution above or below 50%?

The center of the distribution is below 50%

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
#Generating Welch two sample t-test to compare the mean winpercent
#for chocolate and fruity candies
t.test(candy$winpercent[candy$chocolate == 1],
       candy$winpercent[candy$fruity == 1])
```

Welch Two Sample t-test

```
data: candy$winpercent[candy$chocolate == 1] and candy$winpercent[candy$fruity == 1]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
```

```
mean of x mean of y
60.92153 44.11974
```

On average, the chocolate candy has a higher winpercent of 60.92, compared to fruity candy of 44.12.

Q12. Is this difference statistically significant? The p-value of the t=test was 2.871e-08, which is much less than 0.05, therefore the difference is statistically significant.

4. Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
#Sorting dataset by winpercent from lowest to highest and displaying first 5 rows
head(candy[order(candy$winpercent),], 5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

```
#Using dplyr to sort dataset by winpercent from lowest to highest
#and display first 5 rows
candy |>
  arrange(winpercent) |>
  head(5)
```

```
chocolate fruity caramel peanutyalmondy nougat
```

Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The five least liked candy types in this set are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
#Sorting the dataset by winpercent from highest to lowest and displaying first 5 rows
head(candy[order(-candy$winpercent),], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0	1	0	0
Reese's Miniatures	1	0	0	1	0	0
Twix	1	0	1	0	0	0
Kit Kat	1	0	0	0	0	0
Snickers	1	0	1	1	1	1

	crispedricewafer	hard	bar	pluribus	sugarpercent
Reese's Peanut Butter cup	0	0	0	0	0.720
Reese's Miniatures	0	0	0	0	0.034
Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291

Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

```
#Using dplyr to sort dataset by winpercent from highest to lowest
#and display first 5 rows
candy |>
  arrange(desc(winpercent)) |>
  head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisp	rice	wafer	hard bar	pluribus	sugar	percent
Reese's Peanut Butter cup		0	0	0	0		0.720
Reese's Miniatures		0	0	0	0		0.034
Twix		1	0	1	0		0.546
Kit Kat		1	0	1	0		0.313
Snickers		0	0	1	0		0.546

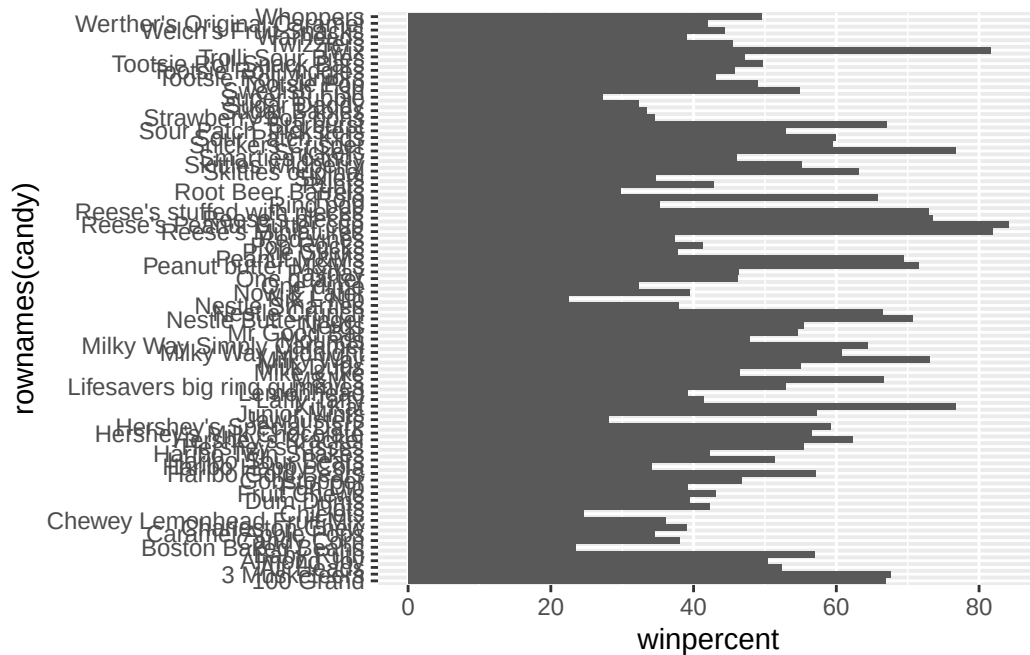
	price	percent	winpercent
Reese's Peanut Butter cup	0.651	84.18029	
Reese's Miniatures	0.279	81.86626	
Twix	0.906	81.64291	
Kit Kat	0.511	76.76860	
Snickers	0.651	76.67378	

The top 5 all time favorite candy types out of this set are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

Overall, I prefer to use the dplyr approach since it seems more organized and easier to read.

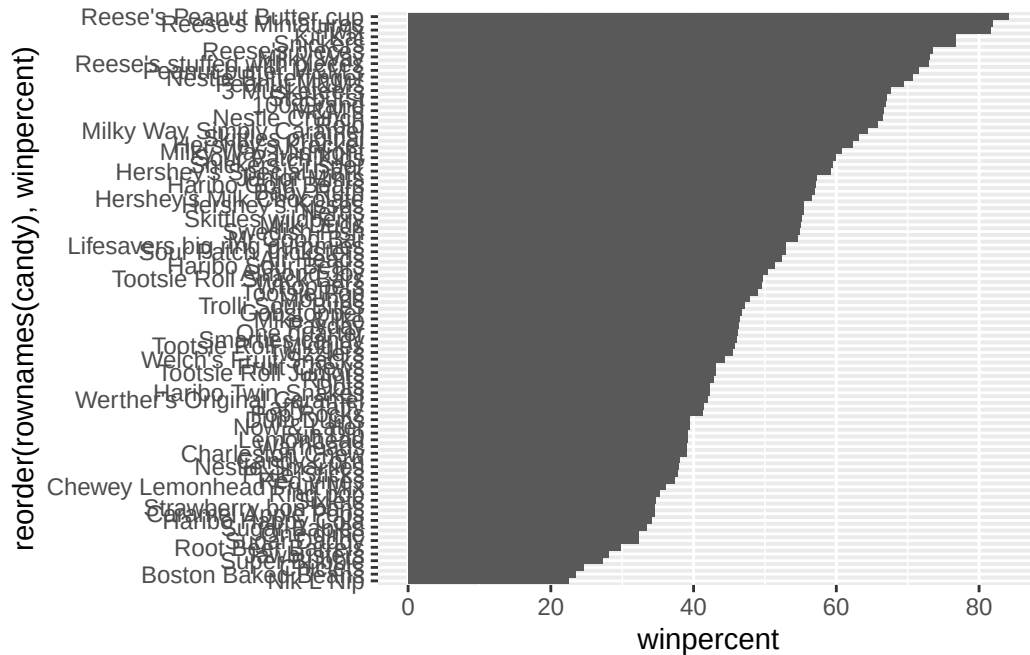
Q15. Make a first barplot of candy ranking based on winpercent values.

```
#Loading ggplot2
library(ggplot2)
#Generating basic barplot of candy ranked based on the winpercent
ggplot(candy) +
  aes(x = winpercent,
      y = rownames(candy)) +
  geom_col()
```



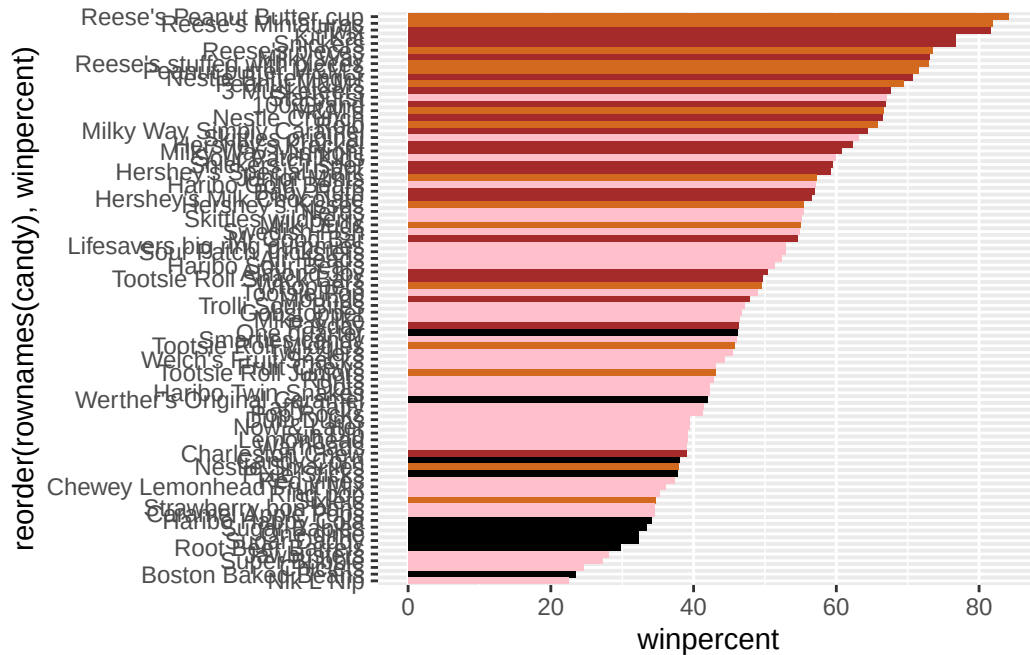
Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
#Loading ggplot2
library(ggplot2)
#Generating barplot with candies sorted by the winpercent
ggplot(candy) +
  aes(x = winpercent,
      y = reorder(rownames(candy), winpercent)) +
  geom_col()
```



4.0.1 Time to add some useful color

```
#Creating vector of colors
#Default all black, one for each candy
my_cols=rep("black", nrow(candy))
#Changing color for candies with chocolate to the color chocolate
my_cols[as.logical(candy$chocolate)] = "chocolate"
#Changing color for candies that are bars to brown
my_cols[as.logical(candy$bar)] = "brown"
#Changing color for candies that are fruity to pink
my_cols[as.logical(candy$fruity)] = "pink"
#Generating barplot with colors based on the type of candy
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

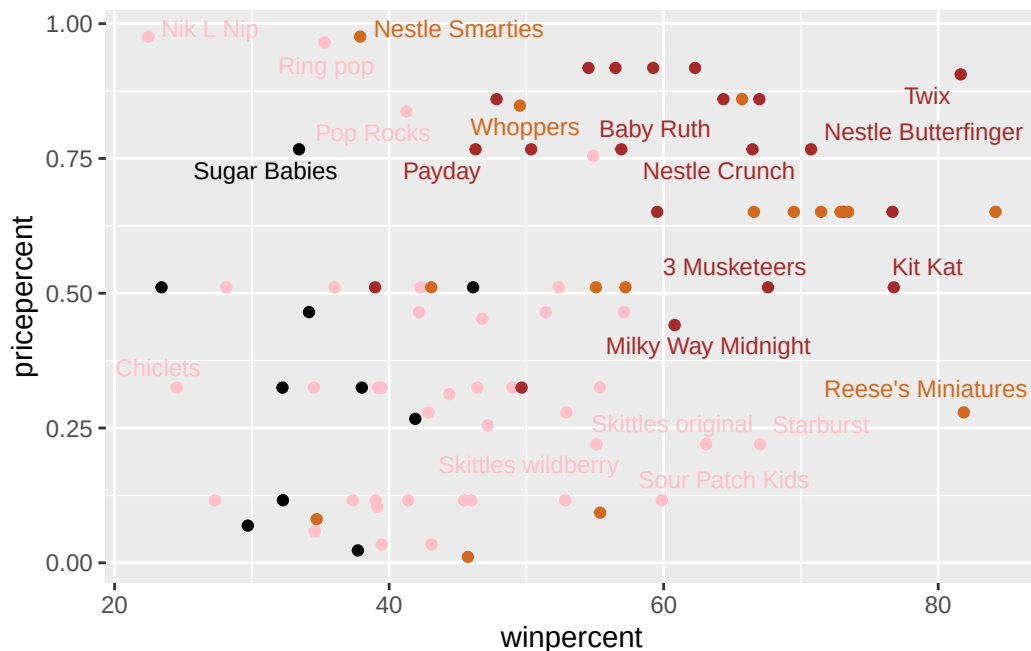
The worst ranked chocolate candy is Sixlets.

Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

5. Taking a look at pricepercent

```
#Loading ggrepel
library(ggrepel)
#Generating plot of winpercent vs pricepercent
ggplot(candy) +
  aes( x = winpercent,
        y = pricepercent,
        label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures is the highest ranked in terms of winpercent for the least money, so it offers the most bang for your buck.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

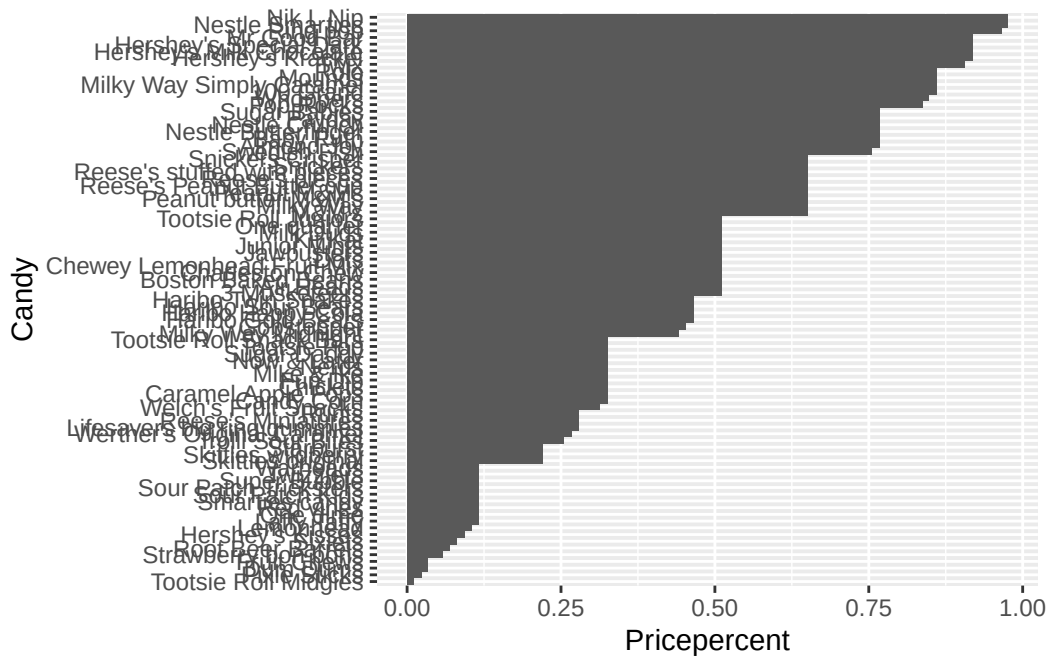
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The five most expensive candies are Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular is the Nik L Nip, since it has the lowest winpercent.

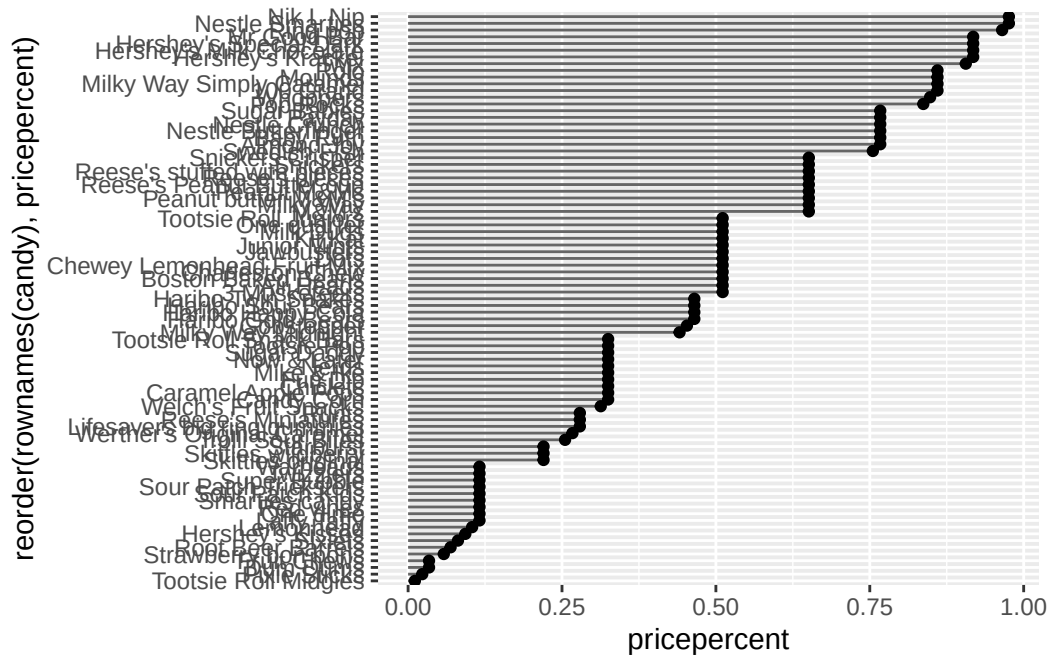
5.1 Optional

Q21. Make a barplot again with `geom_col()` this time using `pricepercent` and then improve this step by step, first ordering the x-axis by value and finally making a so called “dot chat” or “lollipop” chart by swapping `geom_col()` for `geom_point()` + `geom_segment()`.

```
#Generating barplot and order candies by
ggplot(candy)+
  aes(x = pricepercent,
      y = reorder(rownames(candy), pricepercent)) +
  geom_col() +
  labs(y = "Candy", x = "Pricepercent")
```



```
#Generating a lollipop chart of pricepercent
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),
                  xend = 0), col="gray40") +
  geom_point()
```

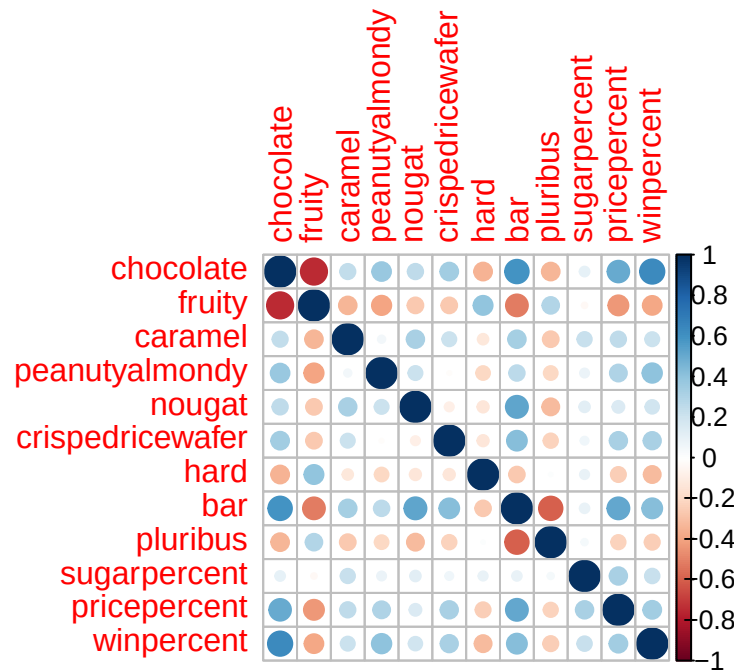


6. Exploring the correlation structure

```
#Loading corrplot package
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
#Generating a correlation matrix of the candy data
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The two variables that are anti-correlated are chocolate and fruity.

Q23. Similarly, what two variables are most positively correlated?

The two variables that are most positively correlated are chocolate and bar.

7. Principal Component Analysis

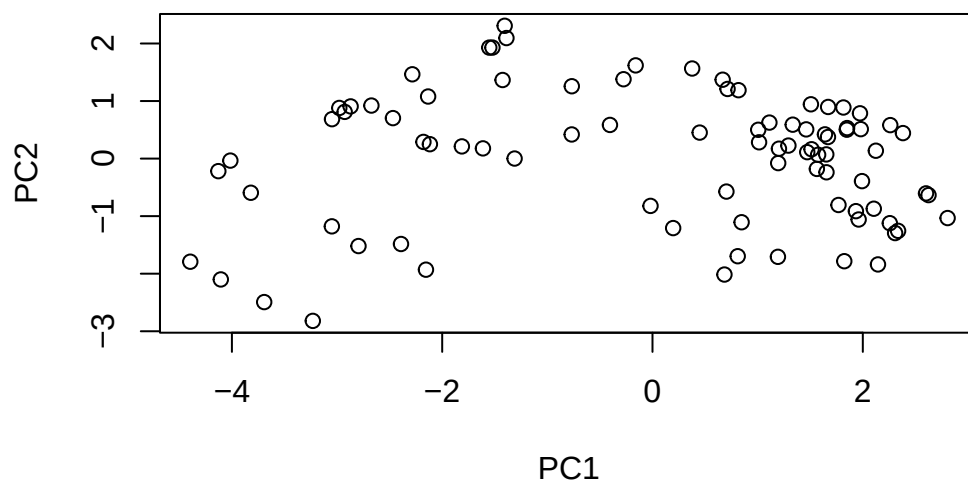
```
#Applying PCA on the candy dataset with scaling
pca <- prcomp(candy, scale = TRUE)
#Viewing summary of PCA results
summary(pca)
```

Importance of components:

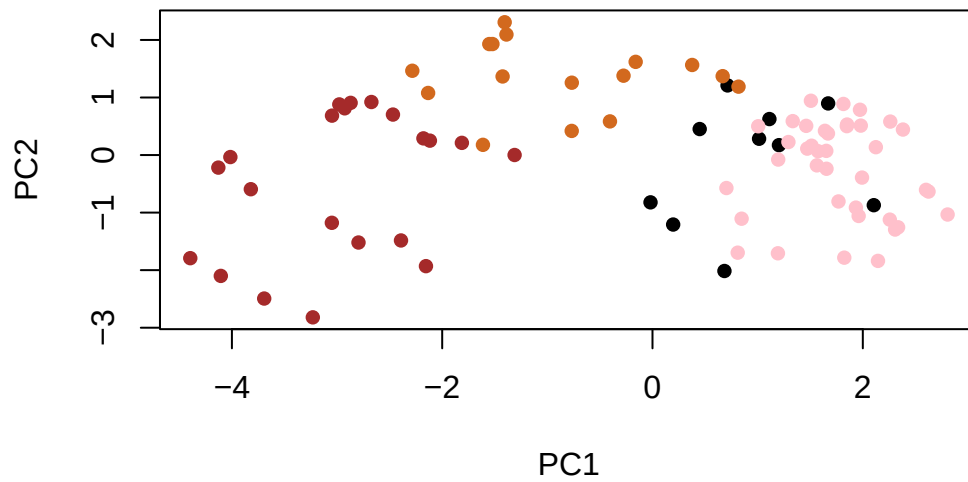
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000


```
#Plotting PC1 vs PC2  
plot(pca$x[,1:2])
```



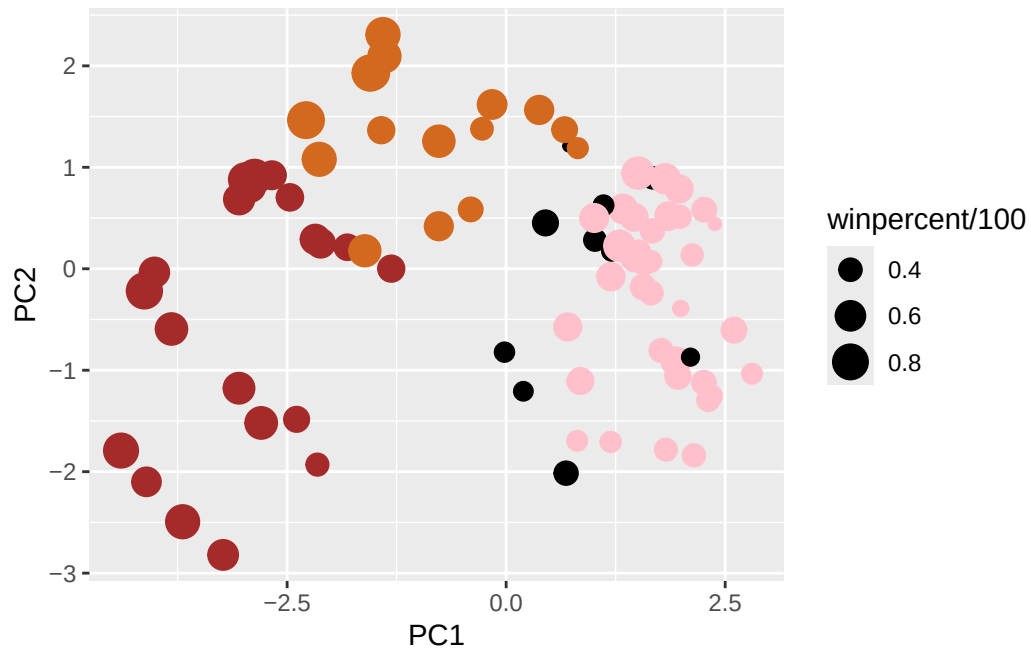
```
#Changing the plotting character and adding color  
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
#Making a new data fram with PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
#Generating PCA plot using ggplot
p <- ggplot(my_data) +
  aes (x = PC1,
        y = PC2,
        size = winpercent/100,
        text = rownames(my_data)) +
  geom_point(col = my_cols)
```

```
#Viewing the plot p
p
```

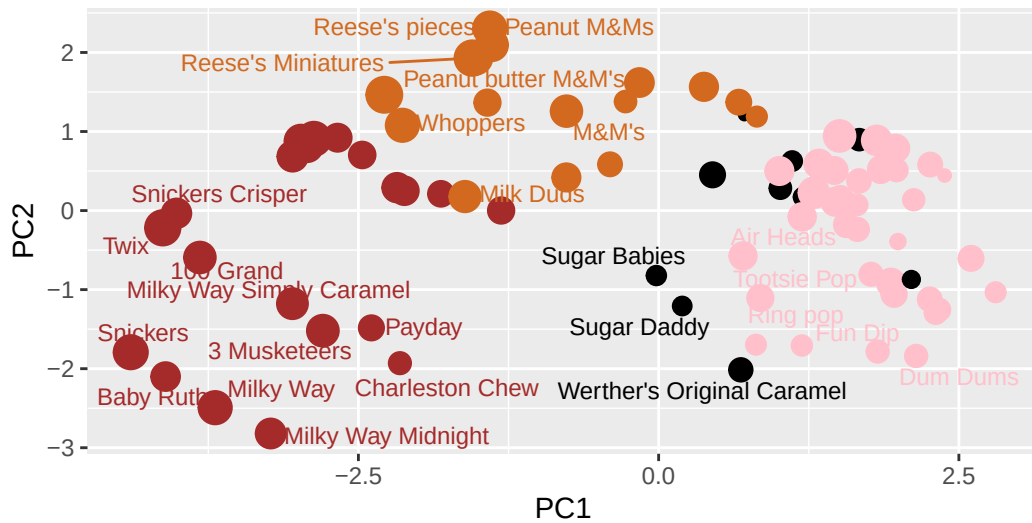


```
#Labeling plot with non overlapping candy names using ggrepel
library(ggrepel)

p +
  geom_text_repel(aes(label = rownames(my_data)),
    size=3.3,
    col=my_cols,
    max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
    subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
    caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
#Using max.overlaps to allow more overlapping lables
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

The following object is masked from 'package:stats':

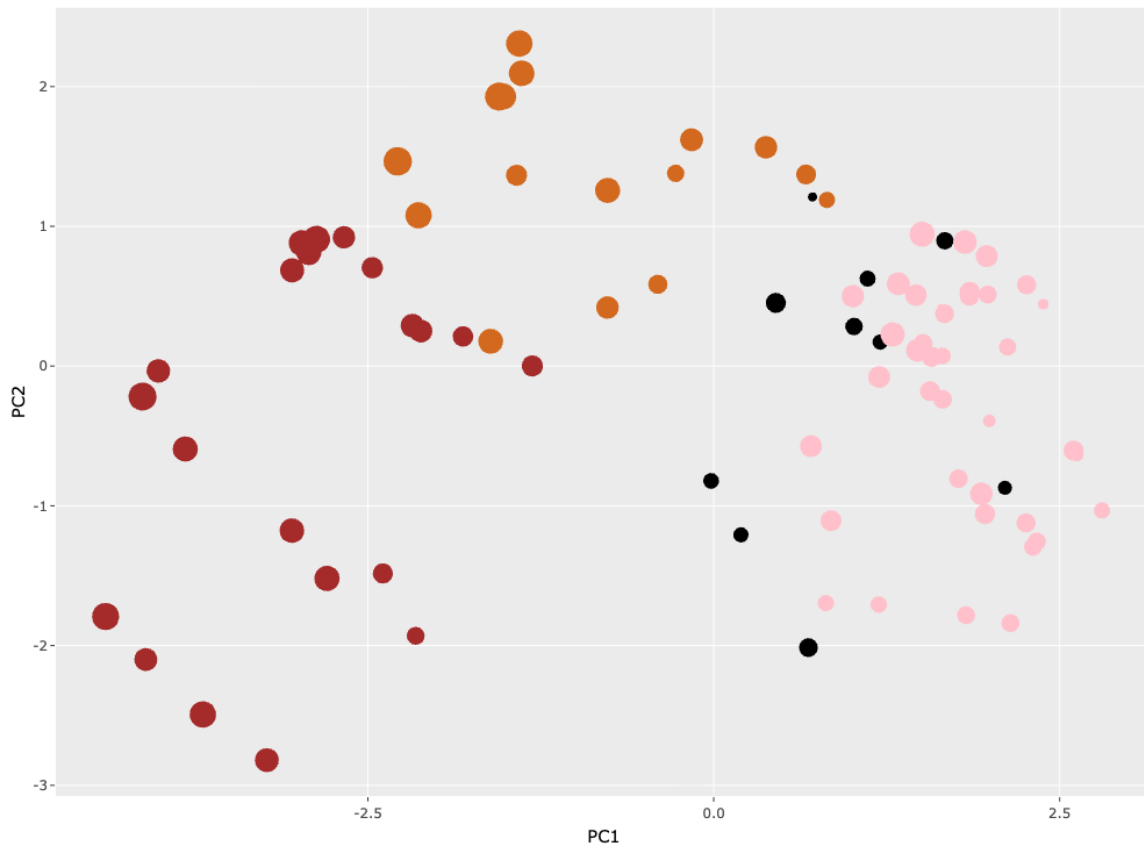
filter

The following object is masked from 'package:graphics':

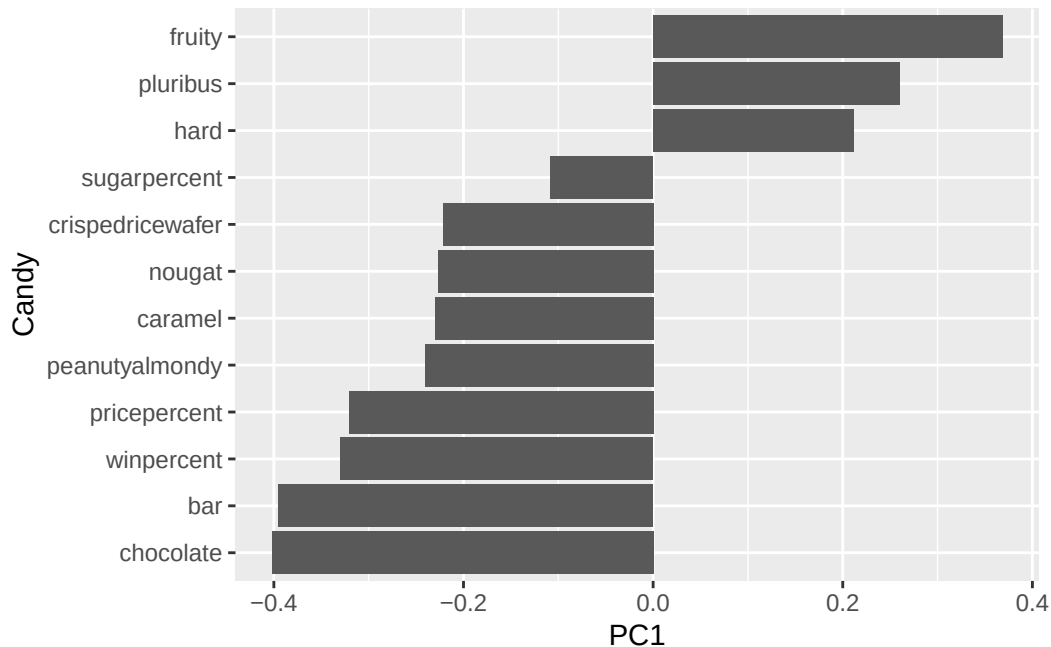
layout

```
library(webshot2)
pp <- ggplotly(p)
htmlwidgets::saveWidget(pp, "plot.html", selfcontained = TRUE)
webshot2::webshot("plot.html", file = "plot.png")
```

file:///Users/leascy/Desktop/BIMM 143/Class 9/Class 9/plot.html screenshot completed



```
#Plotting the loadings for PC1
ggplot(pca$rotation) +
  aes(x = PC1,
      y = reorder(rownames(pca$rotation), PC1)) +
  geom_col() +
  labs (x = "PC1", y = "Candy")
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Original variables that are picked up strongly by PC1 in this positive direction are fruity, pluribus, and hard. This makes sense because these variables describe the features of fruity and non chocolate candies, which are different than chocolate related candies. This relationship was highlighted previously in the correlation plot, since fruity and chocolate were negatively correlated.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy would be a chocolate based candy with added textures such as caramel or peanuts. Visualizations, including bar and scatter plots, highlighted their higher popularity since they have higher winpercent values than fruity candies. Correlation analysis demonstrated how chocolate and fruity candies are strongly anti-correlated, suggesting that these are distinct categories with different popularity levels. PCA further supported this by grouping the most popular candies within the chocolate region.